Taylor & Francis
Taylor & Francis Group

Bayesian Estimation and Prediction for Pareto Data

Author(s): Barry C. Arnold and S. James Press

Source: *Journal of the American Statistical Association*, Dec., 1989, Vol. 84, No. 408 (Dec., 1989), pp. 1079-1084

Published by: Taylor & Francis, Ltd. on behalf of the American Statistical Association

Stable URL: https://www.jstor.org/stable/2290086

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

Taylor & Francis, Ltd. and *American Statistical Association* are collaborating with JSTOR to digitize, preserve and extend access to *Journal of the American Statistical Association*

JSTOR

This content downloaded from
67.186.65.7 on Fri, 30 Apr 2021 16:18:00 UTC
All use subject to https://about.jstor.org/terms

Bayesian Estimation and Prediction for Pareto Data

BARRY C. ARNOLD and S. JAMES PRESS*

Data from a classical Pareto distribution are to be used to make inferences about the inequality and precision parameters. In addition, it is desired to predict the behavior of further observations from the distribution. Three typical data configurations are considered (iid and two types of censoring). Dependent conjugate prior analyses are reviewed and are compared with an analysis involving independent priors for the inequality and precision parameters. It is argued that mathematical tractability should be, perhaps, a minor consideration in the choice of priors. A comparative example is included.

KEY WORDS: Censored data; Conjugate priors; Independent priors.

1. INTRODUCTION

Consider data from a classical Pareto distribution with shape, or inequality, parameter α and precision parameter τ. The corresponding density is

$$f(x;\alpha,\tau) = \tau\alpha(\tau x)^{-(\alpha+1)}, \qquad \tau x > 1$$
$$= 0, \qquad \text{otherwise.} \tag{1.1}$$

The natural parameter space is $\alpha > 0$, $\tau > 0$. If X has density (1.1) we write $X \sim \mathscr{P}(\alpha, \tau)$. If α were known, then a natural conjugate prior density for τ would be a Pareto density; that is,

$$\tau \sim \mathscr{P}(a_0, b_0) \tag{1.2}$$

for some a_0, b_0 chosen to reflect prior knowledge about τ. If τ were known, then the natural conjugate prior family would be the gamma distribution family. Thus we would have

$$\alpha \sim \gamma(c_0, d_0)\left[\text{i.e., } f(\alpha) \propto \alpha^{c_0-1}e^{-d_0\alpha}\right], \tag{1.3}$$

where c_0, d_0 are chosen to reflect prior knowledge about α. If both α and τ are unknown (which is typically the case), a joint prior for (α, τ) must be assessed.

A natural joint conjugate prior for (α, τ) is available. It is intimately related to one first suggested by Lwin (1972) and later generalized in Arnold and Press (1983). The prior will be called a "modified Lwin prior." The form we take may be succinctly described as follows:

$$\alpha \sim \gamma(c_0, d_0)$$

and

$$\tau \mid \alpha \sim \mathscr{P}(\alpha a_0, b_0), \tag{1.4}$$

where a_0, b_0, c_0, and d_0 are positive hyperparameters. For many data configurations from (1.1), the posterior distribution associated with the prior (1.4) is again of the form (1.4) with a_0, b_0, c_0, d_0 replaced by data-dependent quantities a_1, b_1, c_1, and d_1. It is possible to envision how one would assess appropriate values for the hyperparameters in such a distribution. It is not clear, however, why one should accept the implicit dependence between α and τ inherent in such a prior. An alternative joint prior for (α, τ), which can reasonably be accepted and assessed, is one in which α and τ are independent with marginals (1.2) and (1.3). This article is concerned with inference using both of these proposed priors. The material to be presented subsumes and extends work published earlier in Arnold and Press (1983, 1986) and Geisser (1984). Geisser (1984) focused on prediction using diffuse priors. In that article the possible extension to a natural conjugate prior analysis was outlined. Subsequently, in Geisser (1985), a more detailed conjugate prior analysis was presented. Arnold and Press (1983, 1986) focused on estimation. In the Arnold and Press papers, and in that of Lwin (1982), the Pareto distribution was parameterized by shape and scale, instead of by shape and precision. As will be observed, one advantage, first noted by Geisser (1984), of changing to a precision parameterization is that bimodal posterior densities are avoided.

2. POSTERIOR DENSITIES

Although the modified Lwin priors enjoy advantages of mathematical tractability, we have reservations (which we now explain) about accepting the particular dependent structure they involve. Recall that the parameter τ effectively determines the support of the distribution (1.1) and the parameter α determines the shape of the distribution. The prior distribution will summarize *a priori* subjective and/or objective beliefs about the parameters. We contend, along with Geisser (1985, p. 176), that it is more natural to have one's beliefs about the shape depend on one's beliefs about the support rather than vice versa. The modified Lwin prior seems awkward in this context. Of course, if it really does conform with the experimenter's prior assessments it should be used. We believe, however, that the assessment of prior beliefs is most easily achieved marginally, that is, assess beliefs about τ and then, independently, assess beliefs about α. Only if the dependence implicit in (1.4) is a priori acceptable should we try to match the experimenter's prior with a selected distribution of the form (1.4). We focus on and, in the absence of good arguments for (1.4), recommend use of independent priors for α and τ. Following the analysis associated with such priors we will summarize the analogous (more aesthetic) results involving the conjugate modified Lwin priors. We remark that a parallel situation exists when analyzing normal data with unknown mean μ and precision τ. The natural conjugate prior for (μ, τ), as exhibited, for example,

* Barry C. Arnold and S. James Press are Professors, Department of Statistics, University of California, Riverside, CA 92521.

© 1989 American Statistical Association
Journal of the American Statistical Association
December 1989, Vol. 84, No. 408, Theory and Methods

This content downloaded from
67.186.65.7 on Fri, 30 Apr 2021 16:18:00 UTC
All use subject to https://about.jstor.org/terms

by DeGroot (1970, p. 169), involves a curious dependence between μ and τ. In that context (and, more generally, in multivariate normal contexts) Press (1982, p. 254) recommended consideration of independent priors for μ and τ. In Section 5, we present a numerical illustration in which the effects of using both kinds of priors are compared.

2.1 Independent Priors

Suppose that our prior distribution for (α, τ) has independent marginals given by (1.2) and (1.3). We consider three typical data configurations (iid and two types of censoring), and we determine the resulting posterior density for (α, τ) in each case. Whereas the likelihood functions have the same form in the three cases, the parameterizations are different.

2.1.1. iid Observations From the Density (1.1). Denote the realized values by $x_1, x_2, \ldots, x_n$, and let $x_{1:n} \equiv \min(x_1, x_2, \ldots, x_n)$. Using the notation

$$v = \sum_{i=1}^{n} \log x_i$$

we may write the likelihood in the form

$$l(\alpha, \tau) = \alpha^n \tau^{-n\alpha} \exp[-v(\alpha + 1)] I(\tau x_{1:n} > 1)$$

so that the resulting posterior density satisfies

$$f(\alpha, \tau \mid \mathbf{x}) \propto \alpha^{n+c_0-1} \tau^{-(n\alpha+a_0+1)} e^{-(d_0+v)\alpha} \times I(a > 0) I(\tau \min(b_0, x_{1:n}) > 1). \quad (2.1)$$

2.1.2. A Type-I Censored Sample. Here we observe only the k smallest of a sample of size n from the density (1.1). Denote the ordered realized values by $x_{1:n}, x_{2:n}, \ldots, x_{k:n}$. Using the notation

$$w = \sum_{i=1}^{k} \log x_{i:n} + (n - k) \log x_{k:n},$$

we may write the likelihood in the form

$$l(\alpha, \tau) = \alpha^k \tau^{-n\alpha} \exp\left[-w\alpha - \sum_{i=1}^{k} \log x_{i:n}\right] I(\tau x_{1:n} > 1),$$

so the resulting posterior density is such that

$$f(\alpha, \tau \mid \mathbf{x}) \propto \alpha^{k+c_0-1} \tau^{-(n\alpha+a_0+1)} e^{-(d_0+w)\alpha} \times I(\alpha > 0) I(\tau \min(b_0, x_{l:n}) > 1). \quad (2.2)$$

2.1.3. A Type-II Censored Sample. In this case we observe a sample of n independent units only up to time (or level) t. We will observe a random number, say k, of the observations $x_{1:n}, x_{2:n}, \ldots, x_{k:n}$ that are less than t; the others are only known to exceed t in value. Using the notation

$$\tilde{w} = \sum_{i=1}^{k} \log x_{i:n} + (n - k) \log t,$$

we may write the likelihood in the form

$$l(\alpha, \tau) = \alpha^k \tau^{-n\alpha} \exp\left[-\tilde{w}\alpha - \sum_{i=1}^{k} \log x_{i:n}\right] I(\tau x_{l:n} > 1),$$

so the resulting posterior density satisfies

$$f(\alpha, \tau \mid \mathbf{x}) \propto \alpha^{k+c_0-1} \tau^{-(n\alpha+a_0+1)} e^{-(d_0+\tilde{w})\alpha} \times I(\alpha > 0) I(\tau \min(b_0, x_{l:n}) > 1). \quad (2.3)$$

Clearly, although the three cases in (2.1)–(2.3) are sufficiently interesting separately to be set out individually, the three joint posterior densities (2.1)–(2.3) are of the same form. They are members of the family

$$f(\alpha, \tau \mid a, b, c, d, n) \propto \alpha^{c-1} \tau^{-(n\alpha+a+1)} e^{-d\alpha} \times I(\alpha > 0) I(\tau b > 1), \quad (2.4)$$

where a, b, c, and d are positive (but may depend upon n) and n is a natural number. Integration of (2.4) yields marginal posterior densities of the form

$$f(\alpha \mid a, b, c, d, n) \propto (a + n\alpha)^{-1} \alpha^{c-1} e^{-(d-n\log b)\alpha} I(\alpha > 0) \quad (2.5)$$

and

$$f(\tau \mid a, b, c, d, n) \propto \tau^{-(a+1)} (d + n \log \tau)^{-c} I(\tau b > 1). \quad (2.6)$$

The marginal (2.5) was discussed in Arnold and Press (1983). If n is large it approximates a gamma distribution [since for the given data configurations b and d depend on n in a manner such that $d - n \log b \approx d_0 + (n/\alpha)$]. The posterior density for τ, (2.6) is monotone decreasing on its support and has its maximum at $\tau = 1/b$.

The normalizing constant appropriate for the density (2.5) may be evaluated as follows:

$$\int_0^\infty (a + n\alpha)^{-1} \alpha^{c-1} e^{-(d-n\log b)\alpha} d\alpha = n^{-1}(d - n \log b)^{-(c-1)} \int_0^\infty \frac{z^{c-1} e^{-z}}{(z + \rho)} dz,$$

where $\rho = an^{-1}(d - n \log b)$.

The integral appearing in this last expression must be evaluated numerically. For integer values of c it can, by repeated integration by parts, be related to the classical exponential integral function. If c is large, then the integral is of the order of magnitude of $\Gamma(c)$. Observe that

$$\int_0^\infty \frac{z^{c-1} e^{-z}}{z + \rho} dz \le \frac{1}{\rho} \Gamma(c).$$

The ratio of these two quantities will be bounded by 1 and provides a function suitable for tabulation. Thus we define

$$\psi(\eta_1, \eta_2) = \frac{\eta_2}{\Gamma(\eta_1)} \int_0^\infty \frac{z^{\eta_1-1} e^{-z}}{z + \eta_2} dz, \quad (2.7)$$

where η_1 and η_2 are positive. A partial tabulation of this function is provided by Table 1. It is a well-behaved integral, and any reasonable numerical integration technique will be adequate for determination of other values.

This content downloaded from
67.186.65.7 on Fri, 30 Apr 2021 16:18:00 UTC
All use subject to https://about.jstor.org/terms

Table 1. Representative Values of $\psi(\eta_1, \eta_2)$

	η_2							
η_1	2	4	6	8	10	12	14	16
25	.0767	.1421	.1986	.2480	.2915	.3302	.3649	.3956
50	.0392	.0754	.1089	.1400	.1690	.1961	.2214	.2452
75	.0263	.0512	.0749	.0974	.1189	.1393	.1588	.1774
100	.0198	.0388	.0571	.0747	.0917	.1080	.1238	.1390
125	.0159	.0312	.0461	.0606	.0746	.0882	.1014	.1142
150	.0132	.0261	.0387	.0509	.0629	.0745	.0858	.0969
175	.0114	.0225	.0333	.0439	.0543	.0645	.0744	.0842
200	.0099	.0197	.0293	.0386	.0478	.0569	.0657	.0744

NOTE: Refer to Equation (2.7).

In terms of the ψ function we can express the posterior density of α as

$$f(\alpha \mid a, b, c, d, n) = \frac{(1 + (a/n)\alpha)^{-1}\alpha^{c-1}e^{-(d-n\log b)\alpha}I(\alpha > 0)}{(d - n\log b)^{-c}\Gamma(c)\psi(c, (a/n)(d - n\log b))}. \quad (2.8)$$

The mean of the density (2.8) is obtainable by multiplying (2.8) by α and integrating, again using the definition (2.7). This yields

$$E(\alpha \mid a, b, c, d, n) = \frac{c}{(d - n\log b)} \frac{\psi(c + 1, (a/n)(d - n\log b))}{\psi(c, (a/n)(d - n\log b))}. \quad (2.9)$$

Analogous expressions are available for higher posterior moments of α. For any $\xi > -c$, we have

$$E(\alpha^\xi \mid a, b, c, d, n) = \frac{\Gamma(c + \xi)}{\Gamma(c)(d - n\log b)^\xi} \frac{\psi(c + \xi, (a/n)(d - n\log b))}{\psi(c, (a/n)(d - n\log b))}. \quad (2.10)$$

The normalizing constant for (2.6) is expressible as follows:

$$\begin{aligned}\int_{1/b}^{\infty} \tau^{-(a+1)}(d + n\log\tau)^{-c}\,d\tau &= a^{-1}\int_0^{b^a}(d - (n/a)\log z)^{-c}\,dz, \qquad z \equiv \tau^{-a} \\ &= a^{-1}d^{-c}\int_0^{b^a}(1 - (n/ad)\log z)^{-c}\,dz \\ &= a^{-1}d^{-c}\wedge(b^a, n/ad, c), \end{aligned} \quad (2.11)$$

where, for η_1, η_2, and η_3 positive, we define

$$\wedge(\eta_1, \eta_2, \eta_3) = \int_0^{\eta_1}(1 - \eta_2\log z)^{-\eta_3}\,dz. \quad (2.12)$$

Thus

$$f(\tau \mid a, b, c, d, n) = \frac{a\tau^{-(a+1)}(1 + (n/d)\log\tau)^{-c}}{\wedge(b^a, n/ad, c)} I(\tau b > 1). \quad (2.13)$$

It is straightforward to obtain posterior moments of τ in terms of the $\wedge$ function. Thus

$$\begin{aligned}E(\tau^\xi \mid a, b, c, d, n) &= \int_{b^{-1}}^{\infty}\tau^\xi f(\tau \mid a, b, c, d, n)\,d\tau \\ &= \frac{a/(a - \xi)\wedge(b^{a-\xi}, n(a - \xi)d, c)}{\wedge(b^a, n/ad, c)},\end{aligned} \quad (2.14)$$

provided that $\xi < a$.

Triple entry tables of the $\wedge$ function (2.12) (analogous to Table 1) could be prepared by numerical integration. See Arnold and Press (1983) for some related tables [some special cases of the $\wedge(\cdot)$ function are tabulated there, as is a version of the $\psi(\cdot)$ function].

If squared error loss is deemed appropriate, then the posterior means of α and τ [using (2.9) and (2.14)] represent the appropriate Bayes estimators. If there is no compelling reason to accept some specific loss function, then the posterior modes of α and τ may be considered to be satisfactory estimates. As remarked earlier, the posterior density of τ is decreasing on (b^{-1}, ∞) so that the posterior mode is given by b^{-1} [recall that for all three data configurations the value of b in the posterior density is given by $\min(b_0, x_{1:n})$]. The posterior mode of α is a little more troublesome to identify. If we differentiate (with respect to α) the expression on the right side of (2.5) we may verify that $f(\alpha \mid a, b, c, d, n)$ is indeed unimodal (provided that $c > 1$, which will be true for all three data configurations considered). The mode is obtained by iteratively solving

$$\alpha = (c - 1)/[(\alpha + c/n)^{-1} + d - n\log b]. \quad (2.15)$$

If n is large, a satisfactory approximation for this mode is provided by $(c - 2)/[d - n\log b]$.

2.2 Modified Lwin Priors

For comparison purposes we will briefly summarize the analogous results for modified Lwin priors. Much of this material was earlier derived by Geisser (1984, 1985), using a different parameterization. His parameterization assumed that prior information could be subsumed in the form of a sample similar in nature to the one to be collected. For convenience of reference we provide the following list of the relationships between the two parameterizations:

Geisser parameters	*Present parameters*
N_0	a_0
e^{m_0}	b_0
d_0	c_0
$N_0(x_0 - m_0)$	d_0

Using our parameterization, the joint prior for (α, τ) is described by

$$\begin{aligned}\alpha &\sim \gamma(c_0, d_0) \\ \tau \mid \alpha &\sim \mathscr{P}(a_0\alpha, b_0)\end{aligned} \quad (2.16)$$

This content downloaded from
67.186.65.7 on Fri, 30 Apr 2021 16:18:00 UTC
All use subject to https://about.jstor.org/terms

[from (1.4)]. Consequently, we may write

$$f(\alpha, \tau) \propto \alpha^{c_0} e^{-d_0\alpha}(b_0\tau)^{-(\alpha a_0+1)} I(b_0\tau > 1) I(\alpha > 0). \quad (2.17)$$

Consider the three data configurations described in Section 2.1. Refer to Section 2.1.1. For iid observations from (1.1), the resulting posterior density will be

$$f(\alpha, \tau \mid \mathbf{x}) \propto \alpha^{n+c_0} e^{-(d_0+\tilde{v})\alpha}(b_0\tau)^{-((n+a_0)\alpha+1)} I(\alpha > 0) \times I(\tau \min(b_0, x_{1:n}) > 0), \quad (2.18)$$

where

$$\tilde{v} = v + a_0 \log b_0 - (n + a_0)\log[\min(b_0, x_{1:n})], \quad (2.19)$$

in which $v = \sum_{i=1}^{n} \log x_i$.

This is again of the form (2.16). We have

$$\alpha \mid \mathbf{x} \sim \gamma(n + c_0, d_0 + \tilde{v})$$

and

$$\tau \mid \alpha, \mathbf{x} \sim \mathscr{P}((n + a_0)\alpha, \min(b_0, x_{1:n})). \quad (2.20)$$

Analogous expressions are available for the posterior density given the other two data configurations (see Secs. 2.1.2 and 2.1.3). Merely replace the v in (2.19) by w or $\tilde{w}$ from (2.2) or (2.3), and replace $n + c_0$ in (2.20) by $k + c_0$.

The marginal posterior distribution of τ in (2.20) is readily obtained. For convenience define $b_0^* = \min(b_0, x_{1:n})$. We have, for $t > b_0^{*-1}$,

$$\begin{aligned}\Pr(\tau > t \mid \mathbf{x}) &= E(\Pr(\tau > t \mid \mathbf{x}, \alpha) \mid \mathbf{x}) \\ &= E[(t\, b_0^*)^{-(n+a_0)\alpha} \mid \mathbf{x}]\end{aligned}$$

[which is just the moment-generating function of $\alpha \mid \mathbf{x}$ evaluated at $-(n + a_0)\log(t\, b_0^*)$]. Consequently,

$$\Pr(\tau > t \mid \mathbf{x}) = [1 + (n + a_0)\log(t\, b_0^*)/(d_0 + \tilde{v})]^{-(n+c_0)}. \quad (2.21)$$

The posterior density of τ is unimodal with mode at b_0^{*-1}.

Note. If prior information about α or τ is scanty, it may be appropriate to resort to use of a diffuse prior. Diffuseness in our prior for τ, (1.2), can be represented by the (improper) choice $a_0 = 0$, $b_0 = \infty$. Diffuseness in our prior for α, (1.3), can be represented by the (improper) choice $c_0 = d_0 = 0$. There are no particular computational advantages to the choice of a diffuse prior for α. In contrast, a diffuse prior for τ results in considerable simplification. In this case, in our posterior density (2.4) (for any of the three data configurations) we will have $a = 0$ and $b = x_{1:n}$. Consequently, the marginal posterior density of α will be a gamma density. The resulting simplification is especially evident in expressions (2.9), (2.10), (3.7), and (3.8), where all of the ψ's can be deleted.

3. PREDICTION

Suppose that we have collected Pareto data in one of the configurations described in Section 2. Suppose that our prior distribution for (α, σ) has independent marginals given by (1.2) and (1.3). We thus are able to compute a joint posterior density for (α, τ) in the form (2.4) for suitable a, b, c, d, and n. Following Geisser (1984, 1985), we address the question of predicting the behavior of future iid observations $X_1^*, \ldots, X_m^*$ from the density (1.1). The analogous problem of predicting further order statistics based on a Type I censored sample is discussed in the Appendix.

Let $\mathbf{X}^* = (X_1^*, \ldots, X_m^*)$, and let A be a subset of IR_m^+ of interest. We wish to compute

$$\Pr(\mathbf{X}^* \in A \mid \mathbf{x}), \quad (3.1)$$

where $\mathbf{x}$ represents one of the data configurations discussed earlier. In principle, this is readily accomplished by evaluating

$$\Pr(\mathbf{X}^* \in A \mid \alpha, \tau) = \psi_A(\alpha, \tau),$$

say, and then computing

$$\int_0^\infty \int_0^\infty \psi_A(\alpha, \tau) f(\alpha, \tau \mid \mathbf{x})\, d\alpha\, d\tau, \quad (3.2)$$

where the density $f(\alpha, \tau \mid \mathbf{x})$ is the form (2.4) with parameters appropriate for the data configuration $\mathbf{x}$. Needless to say, the computation in (3.2) will be formidable unless A is a "simple" subset of IR_m^+.

We will content ourselves with a single illustrative example. Suppose that $A_{k,m}(w)$ is defined by

$$A_{k,m}(w) = \{\mathbf{x}: \mathbf{x} \in IR_m^+, \text{ exactly } k \text{ of the } x_i\text{'s} > w\}, \quad (3.3)$$

where $w \in IR^+$. Thus we are predicting the proportion of the m future observations that will exceed a specific level w. Evidently, we have

$$\begin{aligned}&\Pr(\mathbf{X}^* \in A_{k,m}(w) \mid \alpha, \tau) \\ &= \binom{m}{k} [\Pr(X_1^* > w \mid \alpha, \tau)]^k [1 - \Pr(X_1^* > w \mid \alpha, \tau)]^{m-k} \\ &= \binom{m}{k} [\min\{(\tau w)^{-\alpha}, 1\}]^k [1 - \min\{(\tau w)^{-\alpha}, 1\}]^{m-k}.\end{aligned} \quad (3.4)$$

Our desired result will be obtained by computing the expected value of (3.4) with respect to the posterior density (2.4). The basic problem involves computing

$$E((\min\{(\tau w)^{-\alpha}, 1\})^l \mid \mathbf{x}), \quad (3.5)$$

since the last term in (3.4) can be replaced by a binomial expansion. Evaluation of (3.5) proceeds as follows [refer to (2.4), (2.7), and (2.8)]:

$$E(\min[(\tau w)^{-\alpha}, 1]^l \mid \mathbf{x}) = \frac{\int_0^\infty \left[\int_{b^{-1}}^\infty \min[(\tau w)^{-l\alpha}, 1]\alpha^{c-1}\tau^{-(n\alpha+a+1)}e^{-d\alpha}d\tau\right]d\alpha}{(n/a)(d - n\log b)^{-c}\psi(c, (a/n)(d - n\log b))\Gamma(c)}. \quad (3.6)$$

This content downloaded from
67.186.65.7 on Fri, 30 Apr 2021 16:18:00 UTC
All use subject to https://about.jstor.org/terms

Case 1: $w > b$.

$$E(\min((\tau w)^{-\alpha}, 1)^l \mid \mathbf{x})$$
$$\propto \int_0^\infty \left[\int_{b^{-1}}^\infty (\tau w)^{-l\alpha}\alpha^{c-1}\tau^{-(n\alpha+a+1)}e^{-d\alpha}\, d\tau\right] d\alpha$$
$$= \int_0^\infty \left[\int_{b^{-1}}^\infty \alpha^{c-1}\tau^{-((l+n)\alpha+a+1)}e^{-(d+l\log w)\alpha}\, d\tau\right] d\alpha.$$

Since this integrand is of the form (2.4), we obtain

$$E(\min((\tau w)^{-\alpha}, 1)^l \mid \mathbf{x}) = \frac{(d + l\log w - (n + l)\log b)^{-c}}{(d - n\log b)^{-c}} \cdot \frac{(n + l)}{n} \times \frac{\psi(c, (a/(n+l))(d + l\log w - (n + l)\log b))}{\psi(c, (a/n)(d - n\log b))}, \tag{3.7}$$

where ψ is as defined in (2.7).

Case 2: $w < b$. By analogous arguments we get

$$E(\min((\tau w)^{-\alpha}, 1)^l \mid \mathbf{x}) = 1 - [(A - B)/C], \tag{3.8}$$

where

$$A = (n/a)\psi(c, (a/n)(d - n\log w))/(d - n\log w)^c,$$
$$B = ((n + l)/a)\psi(c, (a/(n + l))(d - n\log w)) \div (d - n\log w)^c,$$

and

$$C = (n/a)\psi(c, (a/n)(d - n\log b))/(d - n\log b)^c.$$

If modified Lwin priors are deemed appropriate, it is not difficult, using (2.20), to evaluate $\Pr(\mathbf{X}^* \in A_{k,m}(w))$ using expressions analogous to (3.7) and (3.8). [Geisser (1985) provided the details in this case.]

4. ASSESSEMENT OF HYPERPARAMETERS

Individuals generally seem to find it easier to quantify their views about parameters independently of one another, rather than jointly. Then they have to think about only one parameter at a time. If independent marginal priors given by (1.2) and (1.3) are used, then (c_0, d_0) can be assessed to match the experimenter's notion of the location and precision of his prior for α, since

$$E(\alpha) = c_0/d_0$$

and

$$\mathrm{var}(\alpha) = c_0/d_0^2. \tag{4.1}$$

The hyperparameters a_0, b_0 can be assessed by matching two of the experimenter's prior percentiles for τ, with their values given by (1.2). Note that (1.2) implies that

$$\Pr(\tau > t) = (b_0 t)^{-a_0}. \tag{4.2}$$

The experimenter's prior median and 95th percentile for τ will yield two equations that can be solved for a_0 and b_0.

If modified Lwin priors (jointly dependent), (1.4), are deemed appropriate, then we still can assess c_0 and d_0 using (4.1). Prior percentiles for τ may again be used to assess the remaining parameters. In this case, however, we have

$$\Pr(\tau > t) = E(\Pr(\tau > t \mid \alpha)) = [d_0/(d_0 + a_0\log(b_0 t))]^{c_0}. \tag{4.3}$$

If we have already determined appropriate values for c_0 and d_0, then assessment of the experimenter's prior median and 95th percentile for τ will yield, using (4.3), two equations that may be solved for a_0 and b_0.

5. A NUMERICAL EXAMPLE

We will illustrate the effects of the choice of prior family (conjugate or independent marginals) by reanalyzing the Dyer (1981) annual wage data. Annual wage data (in multiples of 100 U.S. dollars) of a random sample of 30 production-line workers in a large industrial firm were as follows: 112, 154, 119, 108, 112, 156, 123, 103, 115, 107, 125, 119, 128, 132, 107, 151, 103, 104, 116, 140, 108, 105, 158, 104, 119, 111, 101, 157, 112, 115. In this case $x_{1:n} = 101$, $n = 30$, and $v = \sum_{i=1}^n \log x_i = 143.523$. Suppose that elicited prior information took the following form: prior mean of $\alpha = 1.9$, prior variance of $\alpha = .6$, prior median for $\tau = 1/100$, and prior 95th percentile for $\tau = 1/85$.

From (4.1) this corresponds to the hyperparameters

$$c_0 = 6.02, \qquad d_0 = 3.17. \tag{5.1}$$

If independent priors are assumed, we solve (4.2) to get

$$a_0 = 14.168, \qquad b_0 = 105.014, \tag{5.2}$$

and the resulting posterior density for (α, τ) [from (2.1)] is given by

$$f(\alpha, \tau \mid \mathbf{x}) \propto \alpha^{35.02}\tau^{-(30\alpha+15.168)} \times e^{-(146.693)\alpha}I(\alpha > 0)I(101\tau > 1). \tag{5.3}$$

If modified Lwin priors are used, then we must solve (4.3) [with c_0 and d_0 given by (5.1)]. In this manner we arrive at

$$a_0 = 10.197, \qquad b_0 = 103.866. \tag{5.4}$$

The resulting posterior density for (α, τ) [from (2.18)] is then of the form

$$f(\alpha, \tau \mid \mathbf{x}) \propto \alpha^{36.02} \times e^{-(8.5247)\alpha}(103.866\tau)^{-[(40.197)\alpha+1]}I(\alpha > 0)I(101\tau > 1). \tag{5.5}$$

The posterior mean of α, using independent priors, is then [from (2.9)] 4.263, and the corresponding value obtained using generalized Lwin priors is 4.225.

The difference is small in this case. It would not, however, be inconsequential if we were predicting extreme future events. Larger differences can be expected if the

This content downloaded from
67.186.65.7 on Fri, 30 Apr 2021 16:18:00 UTC
All use subject to https://about.jstor.org/terms

prior information is more discordant with the data than it was in this case or if the prior precision is high.

APPENDIX: PREDICTION OF FURTHER ORDER STATISTICS

Nigm and Hamdy (1987) considered the problem of predicting $X_{k+1:n}$ based on the order statistics $X_{1:n}, \ldots, X_{k:n}$ from the density (1.1). They used generalized Lwin priors. The analysis can be readily performed for either modified Lwin priors or independent gamma-Pareto priors. The key observation is that, for $t > x_{k:n}$,

$$\Pr(X_{k+1:n} > t \mid x_{1:n}, \ldots, x_{k:n}, \tau, \alpha) = (t/x_{k:n})^{-(n-k)\alpha}. \quad \text{(A.1)}$$

Consequently,

$$\Pr(X_{k+1:n} > t \mid x_{1:n}, \ldots, x_{k:n}) = \int_0^\infty \left(\frac{t}{x_{k:n}}\right)^{-(n-k)\alpha} f(\alpha \mid x_{1:n}, \ldots, x_{k:n})\, d\alpha. \quad \text{(A.2)}$$

The appropriate posterior density $f(\alpha \mid x_{1:n}, \ldots, x_{k:n})$, assuming independent gamma-Pareto priors, is given by (2.5) with hyperparameters determined by (2.2). The resulting expression for (A.2) will be analogous to (3.7).

If instead we assume a modified Lwin prior, then the posterior density of α will be a gamma density [see (2.20) with the appropriate choice of $\tilde{v}$]. Evaluation of (A.2) is then easy, since it can be recognized as the moment-generating function of $\alpha \mid x_{1:n}, \ldots, x_{k:n}$ evaluated at $-(n-k)\log(t/x_{k:n})$.

Specifically, we have in this case

$$\Pr(X_{k+1} > t \mid x_{1:n}, \ldots, x_{k:n}) = \left\{1 + \frac{(n-k)\log(t/x_{k:n})}{d_0 + w + a_0 \log b_0 - (n + a_0)\log[\min(b_0, x_{1:n})]}\right\}^{-(n+c_0)},$$

where $w = \sum_{i=1}^{k} \log x_{i:n} + (n-k)\log x_{k:n}$.

[*Received November 1987. Revised April 1989.*]

REFERENCES

Arnold, B. C., and Press, S. J. (1983), "Bayesian Inference for Pareto Populations," *Journal of Econometrics*, 21, 287–306.

——— (1986), "Bayesian Analysis of Censored or Grouped Data From Pareto Populations," in *Bayesian Inference and Decision Techniques*, eds. P. Goel and A. Zellner, Amsterdam: North-Holland, pp. 157–173.

DeGroot, M. H. (1970), *Optimal Statistical Decisions*, New York: McGraw-Hill.

Dyer, D. (1981), "Structural Probability Bounds for the Strong Pareto Law," *Canadian Journal of Statistics*, 9, 71–77.

Geisser, S. (1984), "Predicting Pareto and Exponential Observables," *Canadian Journal of Statistics*, 12, 143–152.

——— (1985), "Interval Prediction for Pareto and Exponential Observables," *Journal of Econometrics*, 29, 173–185.

Lwin, T. (1972), "Estimation of the Tail of the Paretian Law," *Scandinavian Actuarial Journal*, 55, 170–178.

Nigm, A. M., and Hamdy, H. I. (1987), "Bayesian Prediction Bounds for the Pareto Lifetime Model," *Communications in Statistics*, 16, 1761–1772.

Press, S. J. (1982), *Applied Multivariate Analysis: Using Bayesian and Frequentist Methods of Inference* (2nd ed.), Malabar, FL: Krieger.

This content downloaded from
67.186.65.7 on Fri, 30 Apr 2021 16:18:00 UTC
All use subject to https://about.jstor.org/terms